بِسْمِ اللّٰهِ الرَّحْمٰنِ الرَّحِيْمِ

اَلْحَمْدُ لِلّٰهِ رَبِّ الْعَالَمِيْنَ

وَالصَّلٰوةُ وَالسَّلَامُ عَلٰى خَاتَمِ النَّبِيِّيْنَ

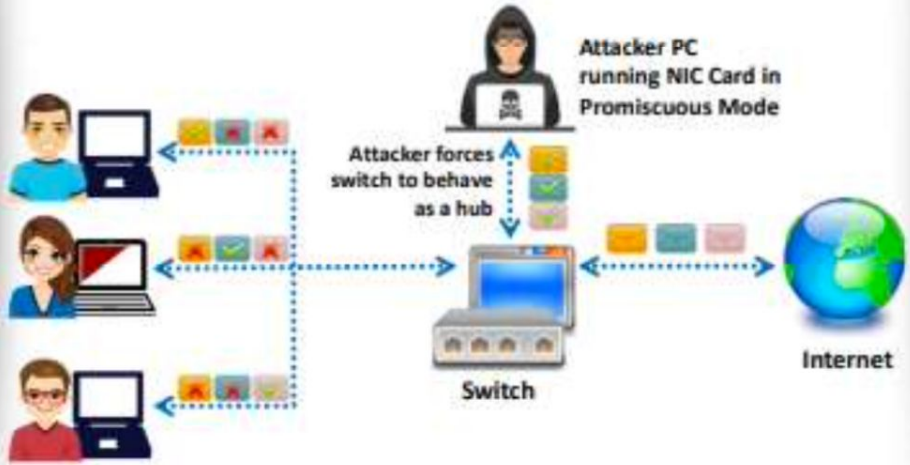
Network Sniffing

Packet Sniffing

- Packet sniffing is the process of **monitoring and capturing all data packets** passing through a given network using a software application or hardware device
- It allows an attacker to observe and **access the entire network traffic** from a given point
- Packet sniffing allows an attacker to **gather sensitive information** such as Telnet passwords, email traffic, syslog traffic, router configuration, web traffic, DNS traffic, FTP passwords, chat sessions, and account information

How a Sniffer Works

- A sniffer turns the NIC of a system to the **promiscuous mode** so that it listens to all the data transmitted on its segment



Sniffing Tools

Promiscuous mode: Configuring a Network Interface Card (NIC) to capture all packets on medium irrespective of the destination hardware (MAC) address

Not easy in Windows and Linux

Kali:

To turn on/off e.g for eth0. Or using Wlan0 e.g. for wifi

sudo ip link set eth0 promisc on

To Check status

ip link show eth0

Packet capture and analysis tools**: Wireshark**

Sniffing from Wired Network

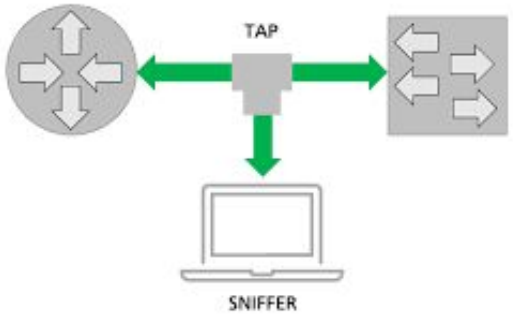
Medium access in Ethernet: Carrier Sense Multiple Access CSMA Collision Detection CD for shared medium

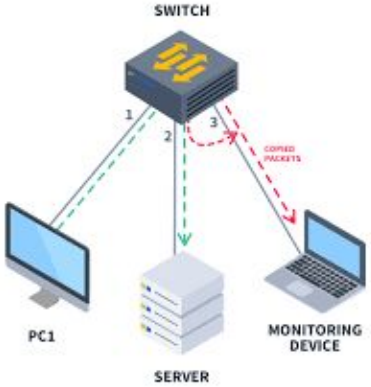
Ethernet Hubs: Flooding at physical layer

Ethernet Switches: MAC based forwarding (except for broadcasts that are flooded)

Port Mirroring: Configuration (router or switch) to copy a source ports traffic to another

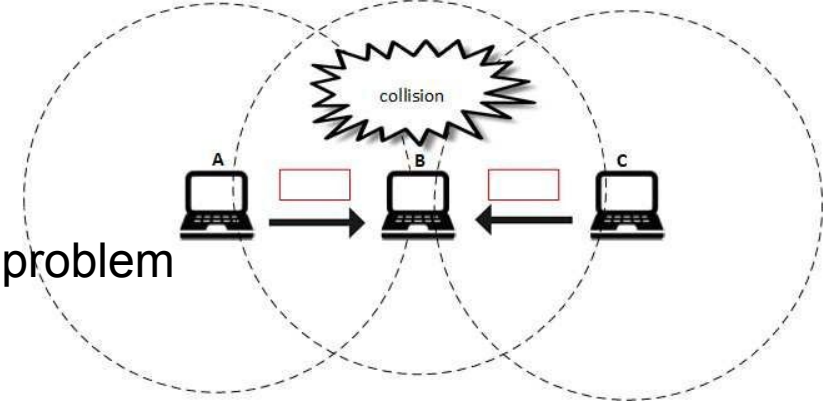
Sniffing on wired Networks: By adding a physical **Tap** or Port mirroring





https://www.endace.com/learn/what-is-packet-sniffing
https://www.cbtnuggets.com/blog/technology/networking/what-is-port-mirroring

Sniffing from Wireless Networks



Medium access in WiFi:

CSMA CA with RTS and CTS due to hidden node problem

Any node in range can capture packets

Hence WiFi needs strong Encryption

WPA Wi-Fi Protected Access protocol using encryption algorithms like AES/RC4 (more on this later)

WEP (Wired Equivalent Privacy) is vulnerable

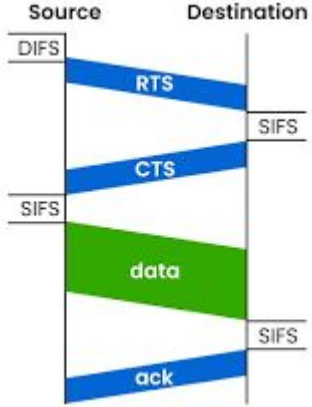
Never share sensitive information on open WiFis

However websites/services can use asymmetric encryption with certificates e.g. HTTPS (more on this later)

CA: Collision Avoidance

RTS: Request to Send

CTS: Clear to Send



CSMA CA

https://doi.org/10.15439/2017F193
https://www.pynetlabs.com/csma-cd-vs-csma-ca/